Alayen Iraqi University
Engineering College/ AI Engineering Department

3rd year

Signals & Systems Lab

Lab (4)

Types of Signals (Discrete)

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2.3 Discrete-Time Signals

discrete-time signals are sequences.

A sequence $x[n]$, $n \in (-\infty, \infty)$ consists of infinite (real or complex) elements or samples.

MATLAB definite-time signals $x[n]$, $n_1 \le n \le n_2$.

1. The definition of the time. For discrete-time signals, time is defined by using *step* 1.
2. The graph of discrete-time signals is obtained by using the `stem` command. `Stem` is similar to `plot` but is suitable for discrete-time signals.

Discrete-Time Signals

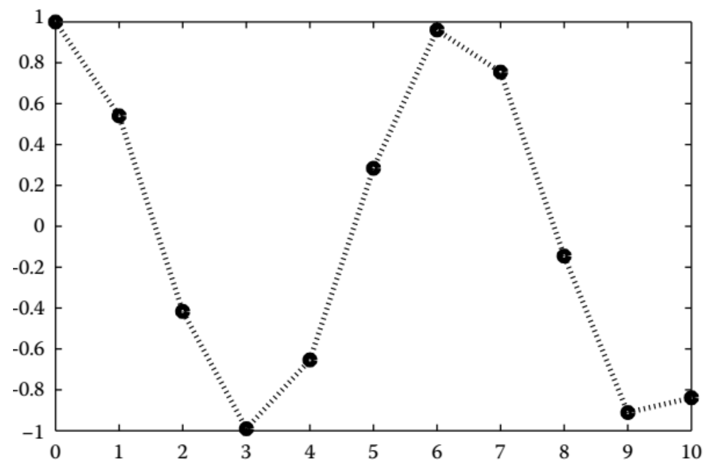
independent variable (time) discrete interval (e.g., the set of integer numbers)

dependent variable continuous set of values

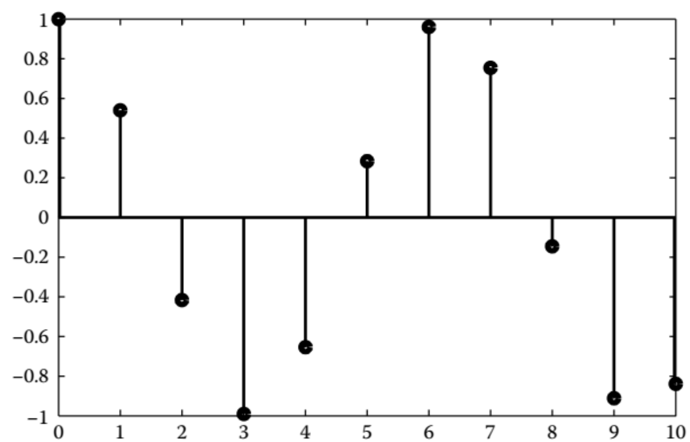
```
n=0:10
```

```
y=cos(n);
```

```
plot(n,y,':o')
```




```
stem(n,y)
```



Discrete time n is defined with step 1.

The dependent variable $y[n]$ is defined in the continuous set of values $-1 \leq y[n] \leq 1$.

Graph of the discrete-time signal $y[n]$ by using the command `plot` with the proper syntax. The points determined by the circles are the values of $y[n]$.

Using the command `stem` is more appropriate when dealing with discrete-time signals.

2.3.1 Unit Impulse Sequence

counterpart of the Dirac delta function

Kronecker delta.

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}.$$

```
n=-3:3
d=gauspuls(n)
```

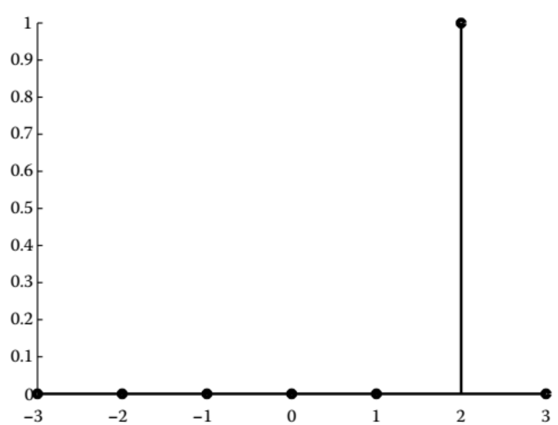
n =	−3	−2	−1	0	1	2	3
d =	0	0	0	1	0	0	0

In the general case

$$\delta[n - n_0] = \begin{cases} 1, & n = n_0 \\ 0, & n \neq n_0 \end{cases}.$$

```
stem(n,d)
```

```
d=gauspuls(n-2);
stem(n,d)
```



Definition and graph of $\delta[n-2]$. The sequence $\delta[n-2]$ is shifted by 2 units to the right compared to $\delta[n]$.

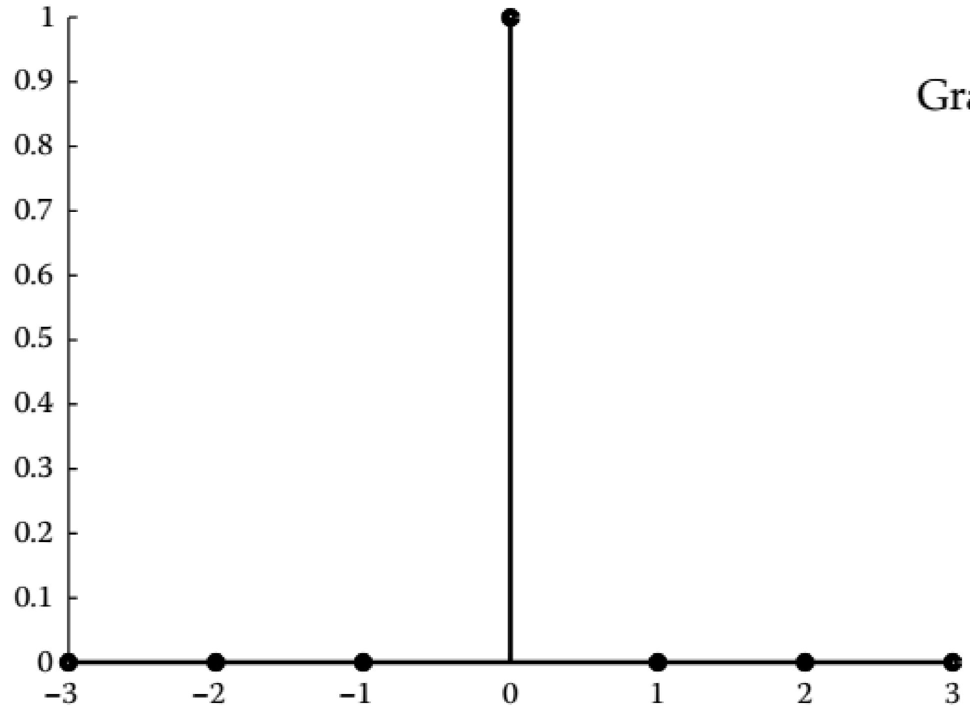
2.3.1 Unit Impulse Sequence

The unit impulse sequence is considered as a three-branched (discrete time) function.

```
n1 = -3:-1;
n2 = 0;
n3 = 1:3;
n = [n1 n2 n3]

d1 = zeros(size(n1));
d2 = 1;
d3 = zeros(size(n3));
d = [d1 d2 d3]

stem(n,d)
```



Graph of $\delta[n]$.

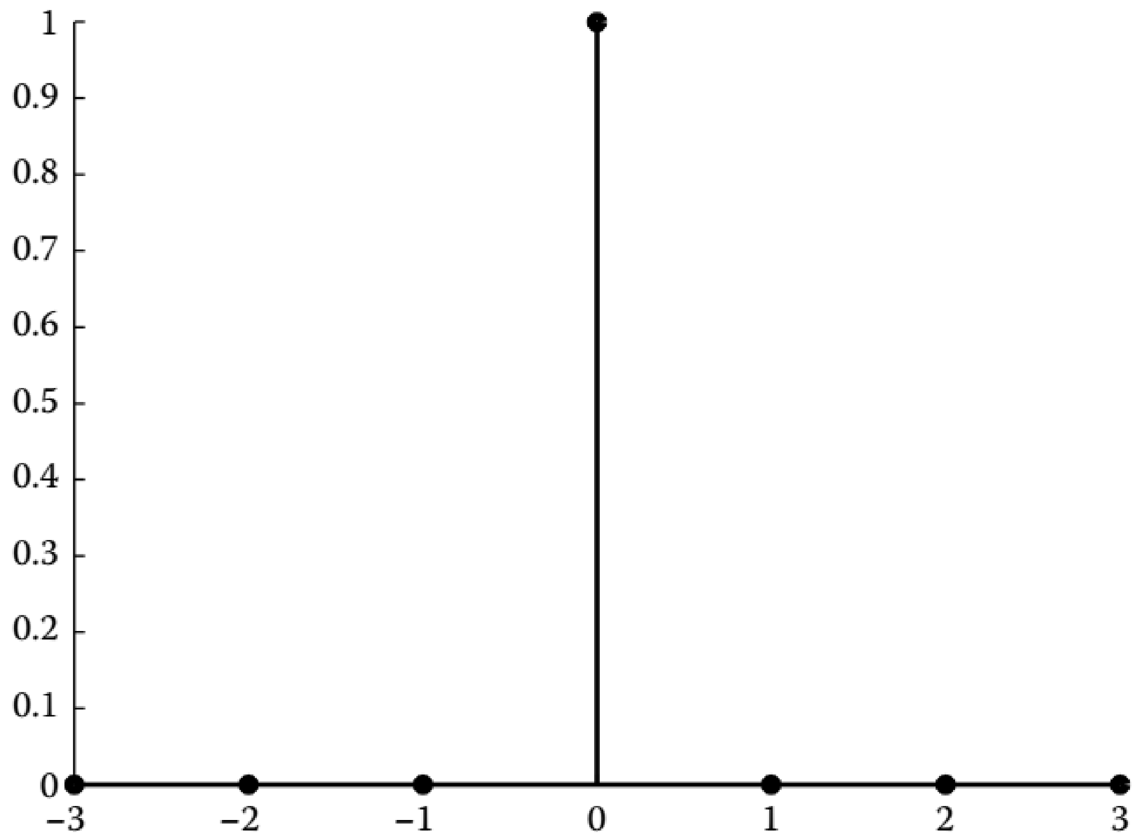
2.3.1 Unit Impulse Sequence

```
n1=-3;
n2=3;
n=n1:n2
d=(n==0)
```

```
n = -3   -2   -1   0   1   2   3

d =  0    0    0   1   0   0   0
```

```
stem(n,d)
```



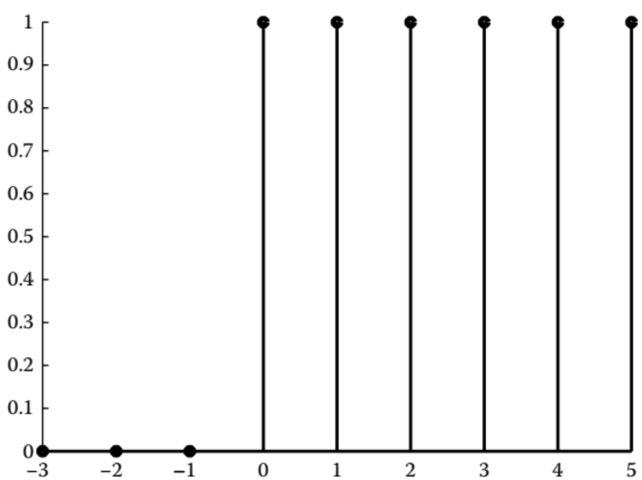
2.3.2 Unit Step Sequence

counterpart of the unit step function

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}.$$

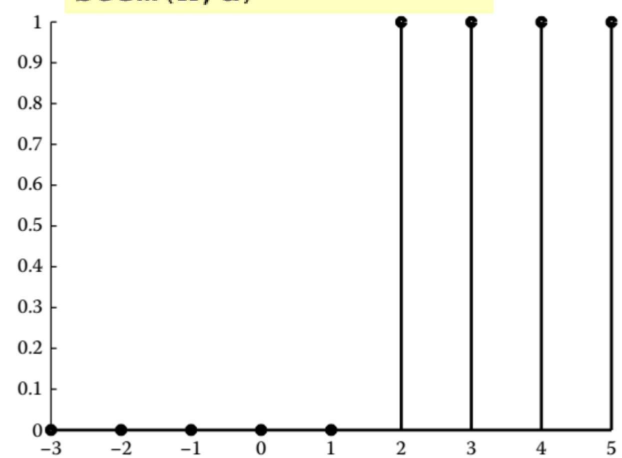
In the general case, $$u[n-n_0] = \begin{cases} 1, & n \geq n_0 \\ 0, & n < n_0 \end{cases}.$$

```
n1=-3:-1;
n2=0:5;
n=[n1 n2];
u1=zeros(size(n1));
u2=ones(size(n2));
u=[u1 u2];
stem(n,u)
```



Graph of $u[n]$ over the (discrete) time interval $-3 \leq n \leq 5$.

```
n1=-3:1;
n2=2:5;
n=[n1 n2];
u1=zeros(size(n1));
u2=ones(size(n2));
u2=ones(size(n2));
u=[u1 u2];
stem(n,u)
```



Graph of $u[n-2]$ over the time interval $-3 \leq n \leq 5$.

2.3.2 Unit Step Sequence

counterpart of the unit step function

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}.$$

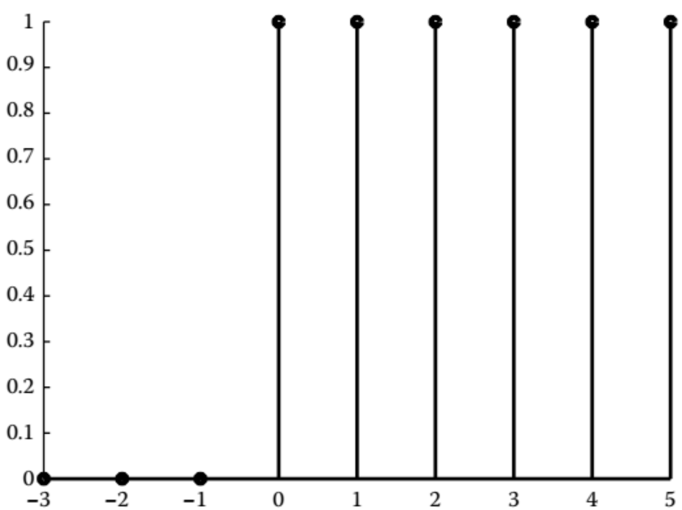
In the general case, $u[n-n_0] = \begin{cases} 1, & n \geq n_0 \\ 0, & n < n_0 \end{cases}.$

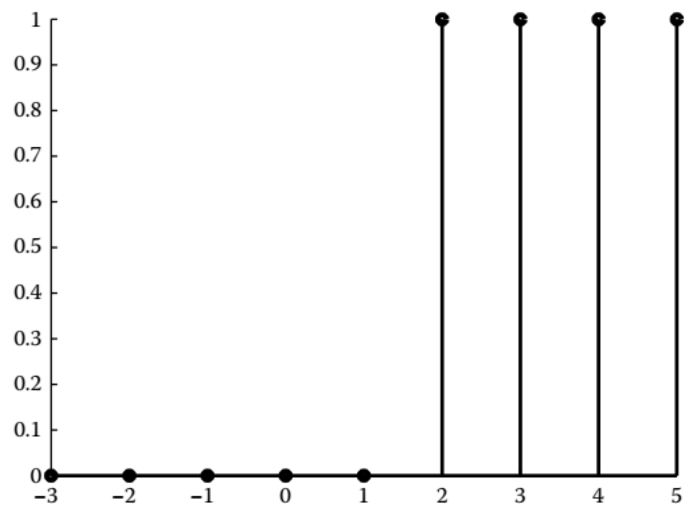
```
n=-3:5
n0=0;
u=(n>=n0)
stem(n,u)
```

Graph of $u[n]$ over the time interval $-3 \leq n \leq 5$.

```
n=-3:5
n0=2;
u=((n-n0)>=0)
stem(n,u)
```

Graph of $u[n-2]$ over the time interval $-3 \leq n \leq 5$.





2.3.3 Real Exponential Sequence

real-valued exponential sequence

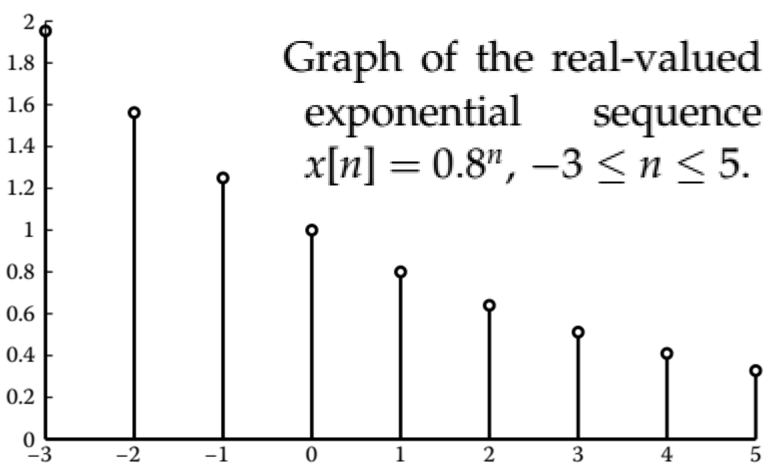
$x[n] = a^n, a \in \mathbb{R}$.

if $|a| > 1$ ascending

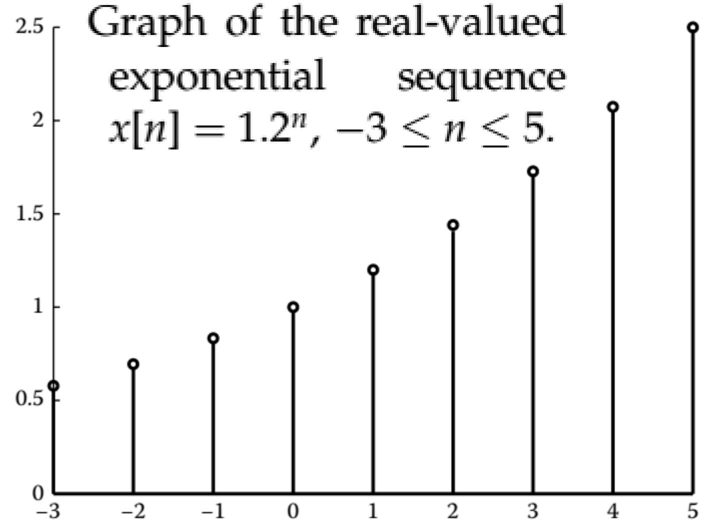
$|a| < 1$ descending

```
n=-3:5
a1=0.8;
x1=a1.^n;
stem(n,x1);
```

```
a2=1.2;
x2=a2.^n;
stem(n,x2);
```



Graph of the real-valued exponential sequence $x[n] = 0.8^n, -3 \le n \le 5$.



Graph of the real-valued exponential sequence $x[n] = 1.2^n, -3 \le n \le 5$.

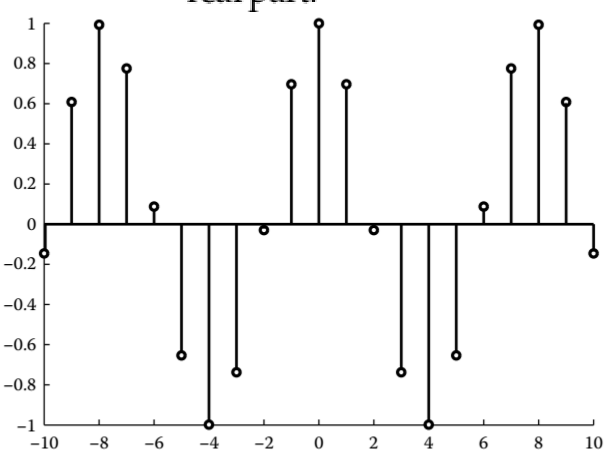
2.3.4 Complex Exponential Sequence

$x[n] = e^{j\omega n}$ Euler's formula

$x[n] = \cos(\omega n) + j\sin(\omega n).$

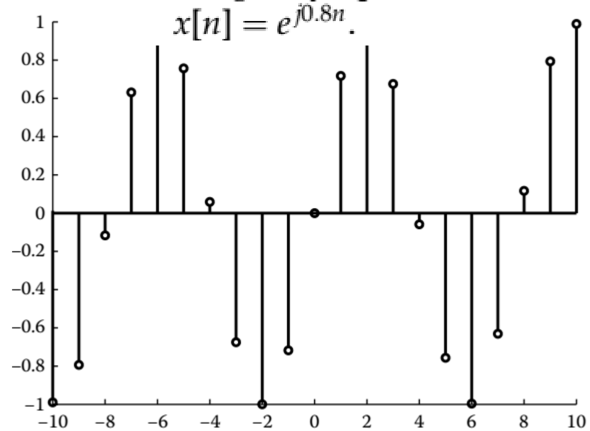
```
n=-10:10;
w=0.8;
x=exp(j*w*n);
stem(n,real(x));
```

Definition of the complex exponential sequence $x[n] = e^{j0.8n}$, $-10 \le n \le 10$ and graph of its real part.



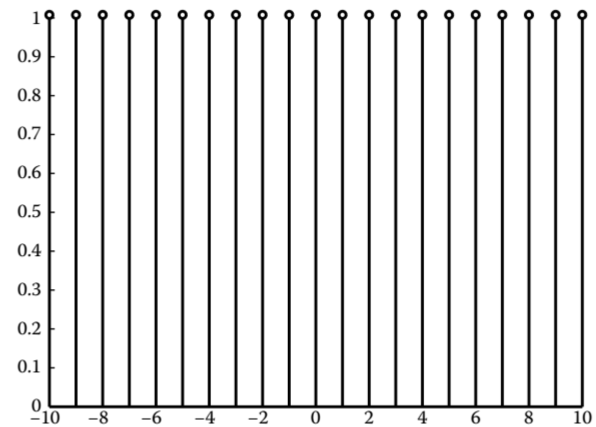
```
stem(n,imag(x))
```

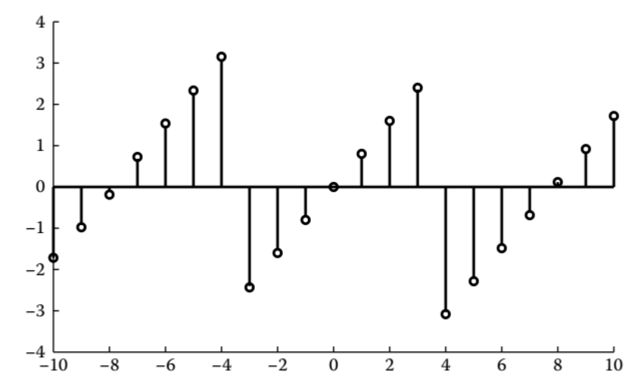
Graph of the imaginary part of $x[n] = e^{j0.8n}$.



```
stem(n,abs(x))
```

Graph of the magnitude of $x[n] = e^{j0.8n}$.





```
stem(n,angle(x))
```

Graph of the angle of $x[n] = e^{j0.8n}$.

2.3.4 Complex Exponential Sequence

$x[n] = e^{j\omega n}$ Euler's formula

$x[n] = \cos(\omega n) + j\sin(\omega n).$

A more general form for a complex exponential sequence

$x[n] = r^n e^{j\omega n}$

$x[n] = z^n, \quad z = re^{j\omega}$

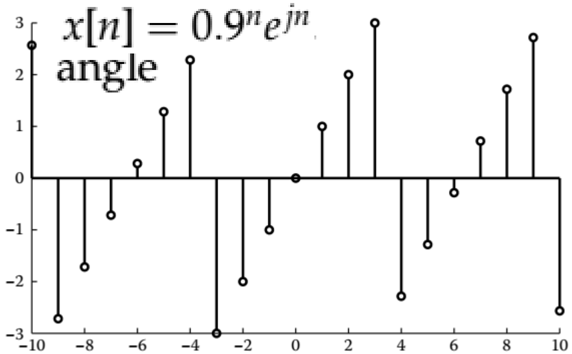
```
z=0.9*exp(j*1)
```

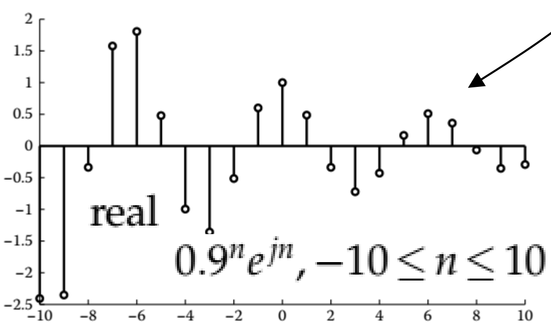
```
z = 0.4863+0.7573i
```

```
n=-10:10;
x=z.^n;
stem(n,real(x));
stem(n,imag(x));
```

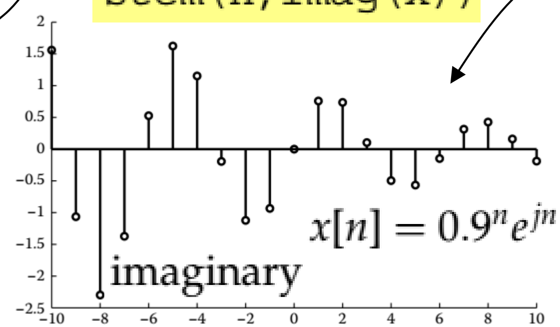
```
n=-10:10;
r=0.9;
w=1;
x=(r.^n).*exp(j*w*n);
stem(n,real(x));
```

```
stem(n,angle(x))
```

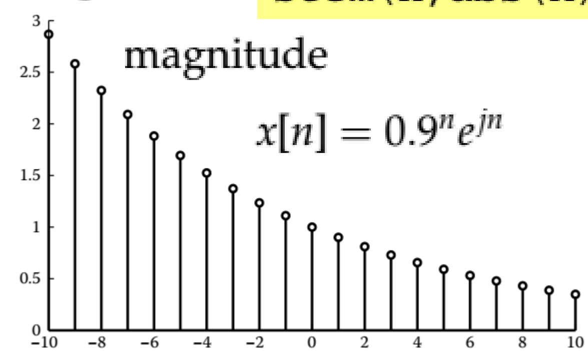





```
stem(n,imag(x))
```




```
stem(n,abs(x))
```



2.3.5 Sinusoidal Sequence

$x[n] = A\cos(\omega n + \varphi)$

or

$x[n] = A\sin(\omega n + \varphi)$

A amplitude

φ phase

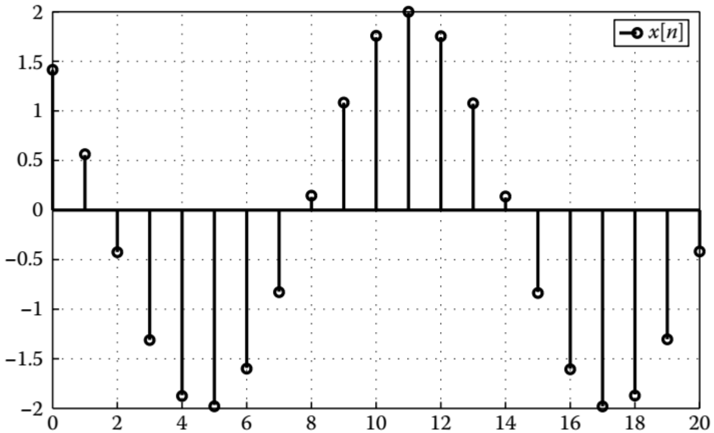
ω angular frequency

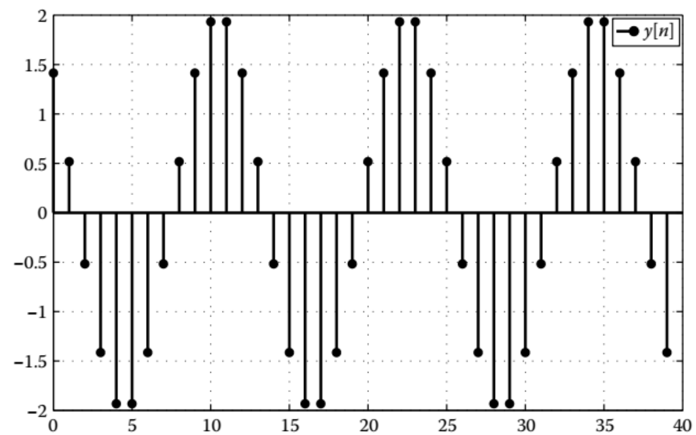
Example

Plot the sinusoidal sequences $x[n] = 2\cos(n/2 + \pi/4)$, $0 \le n \le 20$ and $y[n] = 2\cos(n\pi/6 + \pi/4)$ $y[n] = 2\cos(n\pi/6 + \pi/4)$, $0 \le n \le 20$.

```
n=0:20;
x=2*cos(1/2*n+pi/4);
stem(n,x)
legend('x[n]')
grid
```

```
y=2*cos(pi/6*n+pi/4);
stem(n,y)
legend('y[n]')
grid
```





2.3.5 Sinusoidal Sequence

$x[n] = A\cos(\omega n + \varphi)$

or

$x[n] = A\sin(\omega n + \varphi)$

A amplitude

φ phase

ω angular frequency

$x(t) = \cos(\Omega t) \Rightarrow x[n] = \cos(\omega n) = \cos(\omega n T_s).$

sampling rate $f_s = 1/T_s$

$f = \Omega/2\pi$

Nyquist–Shannon sampling theorem

$$f_s > 2f. \Rightarrow T_s < \frac{1}{2f} = \frac{\pi}{\Omega}$$

2.3.5 Sinusoidal Sequence

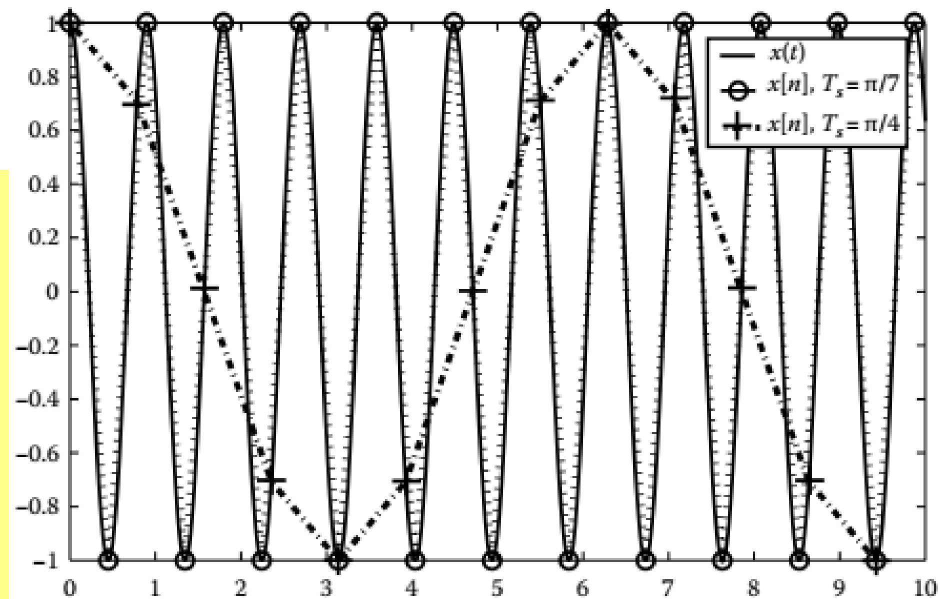
$x(t) = \cos(\Omega t) \Rightarrow x[n] = \cos(\omega n) = \cos(\omega n T_s)$. sampling rate $f_s = 1/T_s$

Nyquist–Shannon sampling theorem $f = \Omega/2\pi$ $f_s > 2f. \Rightarrow T_s < \dfrac{1}{2f} = \dfrac{\pi}{\Omega}$

$x(t) = \cos(7t) \Rightarrow x[nT_s]$

$T_s = \pi/4 \Rightarrow T_s > \pi/\Omega$ $f_s = 1/T_s = 4/\pi < 2f = 2(\Omega/2\pi) = 7/\pi$

$T_s = \pi/7 \Rightarrow T_s \leq \pi/\Omega = \pi/7$




```
t=0:0.01:10;
x=cos(7*t);
Ts1=pi/7;
ts1=0:Ts1:10;
xs1=cos(7*ts1);
Ts2=pi/4;
ts2=0:Ts2:10;
xs2=cos(7*ts2);
plot(t,x,ts1,xs1,':o',ts2, xs2,':+')
legend('x(t)','x[n],T_s=\pi/7','x[n],T_s= \pi/4')
```


Thank
you!